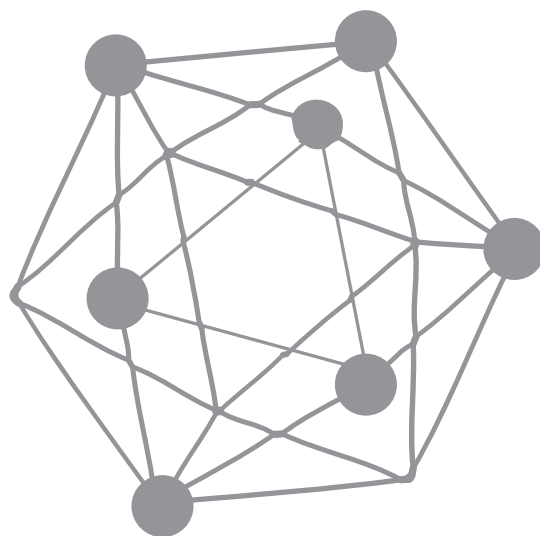


Hyperledger Fabric: What You Should Know and Why



Understanding the essential features of Hyperledger Fabric is crucial for anyone looking to develop blockchain applications for enterprise use. These features provide the foundation for building secure, scalable, and privacy-focused applications, and can be leveraged to create innovative solutions that address real-world business challenges.

Blockchain technology has rapidly gained popularity across various industries for its secure and transparent nature. Hyperledger Fabric, one of the most prominent blockchain frameworks, stands out for its robust features and flexibility. Designed for enterprise use, it is part of the Hyperledger project hosted by The Linux Foundation and is one of the most popular blockchain frameworks for developing business applications.

How does it work?

At its core, Hyperledger Fabric works by creating a network of unique organisations that interact with each other. Each organisation has its own Fabric certificate authority (CA) and one or more peer nodes. The Fabric CA issues certificates for users and manages permissions, while peer nodes endorse transactions, execute smart contracts (Chaincode), and store a local copy of the ledger.

Transactions in Hyperledger Fabric are initiated by clients and endorsed by endorsing peers. These endorsed transactions are then ordered into blocks by the ordering service, which is shared by all organisations in the network. The ordering service ensures that transactions are properly ordered and broadcasts new blocks to peer nodes. Peer nodes validate the transactions in the block and update their local copy of the ledger accordingly.

Hyperledger Fabric as a blockchain framework

Hyperledger Fabric is a blockchain framework designed for developing enterprise-grade applications. It provides a modular architecture that allows organisations to

customise various components, such as consensus algorithms, membership services, and smart contract execution environments.

Fabric's key features include permissioned network, privacy and confidentiality, scalability, smart contracts (Chaincode), endorsement policies, pluggable consensus, identity management, rich querying, and interoperability. These features make Fabric well-suited for building secure and scalable blockchain applications for enterprise use.

The essential features of Hyperledger Fabric

Here's a detailed overview of each feature of Hyperledger Fabric.

Modularity: Hyperledger Fabric's modular architecture allows for flexibility and customisation by breaking down the blockchain network into independent components. These components, such as the ordering service, peer nodes, and membership service provider (MSP), can be customised or replaced to meet specific business requirements. For example, organisations can integrate their consensus algorithms or use different databases for storing ledger data, enhancing the network's scalability and performance.

Permissioned network: Hyperledger Fabric operates on a permissioned network, where participants must be authenticated and authorised to access the network. This enhances security and privacy by ensuring that only trusted parties can participate in the network. Permissioned networks also enable organisations to comply with regulatory requirements by controlling access to sensitive information.